Ankush Reddy Sugureddy

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SUMMARY

A highly accomplished Lead Engineer and Architect with over 15 years of experience in designing, building, and optimizing petabyte-scale, multi-cloud data platforms (GCP, AWS, Azure). Specializing in cutting-edge Generative AI and Agentic AI frameworks, including leading LLMOps strategy (Kimi, Llama) and integrating GenAI-powered BI solutions. Proven expertise across the modern data stack (Spark, Python, BigQuery, dbt, Kubernetes, Terraform) to deliver high-performance, real-time data solutions, drive automated data quality/governance, and significantly boost enterprise workflow efficiency.

EXPERIENCE

Lead Engineer | Cloudflare **11/2022 to Present**

- Generative AI & LLMOps Strategy: Spearheaded enterprise-wide AI strategy, deploying LLM pipelines using Kimi and Llama foundation models to generate automated, context-aware insights on IT operational data (Salesforce, Workday, JIRA), significantly boosting workflow efficiency.
- GenAI-Powered BI & Analytics: Architected and integrated a Chat and Personal Intelligence layer into Apache Superset; enabled dashboard-aware Q&A, persistent user sessions, and chart-specific data selection, drastically reducing time-to-insight for analysts and minimizing ad-hoc SQL dependencies.
- Petabyte-Scale Data Architecture: Managed petabyte-scale data processing across a hybrid and multi-cloud ecosystem (GCP, Cloudflare Dash, On-Prem) for billions of Stripe billing records and enterprise domains (Salesforce, NetSuite, Workday, JIRA, Greenhouse, Postgres).
- Real-Time Streaming APIs: Engineered high-performance, real-time API integrations with Salesforce to process compute-intensive financial metrics (ACV, Deferred ACV, Accumulated Deferred ACV); achieved an end-to-end processing latency of under 30 seconds per request.
- Modern Medallion Lakehouse: Designed and deployed an end-to-end Medallion Architecture (Landing, Refined, Transformed), enforcing robust data governance through automated lineage tracking, data classification, and advanced data hashing/encryption.
- Automated Data Quality & Governance: Championed the adoption of Soda Core for comprehensive, automated data quality testing and implemented robust security standards across the multi-cloud data platform.
- Modern Data Stack & CI/CD: Implemented dbt within CI/CD pipelines for BigQuery transformations, establishing data quality testing, modularizing financial reporting models, and automating upstream/downstream impact analysis using dbt artifacts.
- Infrastructure & Reliability: Managed GCP resources and infrastructure as code (IaC) using Terraform for consistency, reproducibility, and high system reliability across Kubernetes clusters.
- Environment / Tech Stack: GCP, Hybrid/Multi-Cloud, Python, Spark, BigQuery, dbt, Apache Superset, Kimi, Llama, Soda Core, APIs, Kubernetes, Terraform.

SDE/ML Engineer | Amazon **07/2021 to 11/2022**

- Led end-to-end migration of legacy on-prem data workflows (Oracle, IBM Cognos, DRM) to AWS, achieving 100% data fidelity and seamless processing of complex enterprise data.
- Designed and maintained scalable ingestion pipelines using Python/Spark on EMR, orchestrating complex workflows with AWS Managed Airflow and custom operators to support 10+ enterprise domains.
- Implemented low-latency, real-time streaming data ingestion using Kinesis and SQS, ensuring sub-minute data availability for critical downstream applications.
- Architected and migrated data platform to a three-layer Medallion Lakehouse model (S3/Glue Data Catalog), improving data governance and reducing query costs by 20%.
- Utilized dbt for complex data modeling and transformations within the Lakehouse, enforcing high data quality and reducing transformation failures by 15%.
- Automated ML pipelines, data lineage, and validation frameworks, accelerating feature engineering time by 30% and supporting 5+ Data Science teams.
- Automated data transfer and triggered Airflow pipelines based on S3 events using AWS Lambda functions, increasing pipeline reactivity.
- Developed and optimized scalable end-to-end ingestion pipelines for TB-scale enterprise data using Python/Spark and micro-batching, guaranteeing low-latency data availability.
- Engineered and deployed high-impact data solutions using robust CI/CD processes, leveraging AWS Managed Airflow and IaC principles for 99.9% system reliability.
- Environment / Tech Stack: AWS, Spark, Python, Scala, EMR, Airflow, Kinesis, SQS, S3, Glue, dbt, Lambda.

Machine Learning Engineer | Walmart **01/2020 to 07/2021**

- Led Machine Learning Engineering initiatives, driving a 12% improvement in ML model performance and developing robust, high-availability production pipelines.
- Developed advanced forecasting models (FBProphet, SARIMA, XGBoost) to predict on-hand inventory across all 4,700+ stores, resulting in a 10% reduction in stockouts.
- Leveraged BigQuery for complex joins and transformations during the feature engineering stage, processing over 50TB of raw data.
- Utilized Dataproc and PySpark on GCP to execute distributed training and forecasting of ML models at scale, reducing model training time by 40%.
- Integrated Snowflake as the primary staging/consumption layer, optimizing query performance for ML feature stores and model serving by up to 65%.
- Introduced dbt governance to modularize complex SQL transformations in the forecasting data pipeline, accelerating feature deployment by 25% and ensuring high data integrity.
- Standardized ETL processes to automate data extraction and manual reporting, resulting in a 30% reduction in operational manual hours across the team.
- Built and deployed ML pipelines on Azure Databricks, utilizing Spark for store-level aggregations and loading resultant data into CosmosDB for low-latency serving.
- Environment / Tech Stack: GCP, BigQuery, Dataproc, PySpark, Python, Snowflake, dbt, Azure Databricks, CosmosDB, FBProphet, XGBoost.

Big Data Architect | Revlon **06/2018 to 01/2020**

- Designed and implemented robust ETL processes for high-volume POS data migration to Azure SQL DW, leveraging Logic Apps for workflow automation and achieving 99.5% uptime.
- Conducted successful PoC for Snowflake integration, building Spark applications to facilitate seamless data exchange between Snowflake and Azure environments.
- Designed the target-state Snowflake data warehouse architecture, driving the migration of multi-terabyte legacy data and achieving a 40% reduction in query latency.
- Pioneered the introduction of dbt for version control and management of SQL transformations in Azure SQL DW, significantly improving data reliability.
- Standardized the transformation layer using dbt modular data modeling, and integrated automated data quality checks via dbt tests for all production pipelines, catching 95% of data anomalies before production.
- Implemented high-throughput data processing using Azure Databricks and Spark Streaming/PySpark to support real-time applications with sub-second latency.
- Developed an interactive offer simulator using R-Shiny to calculate Return on Investment (ROI) and optimize marketing offer calendar planning, leading to a 15% lift in campaign efficiency.
- Applied PySpark NLP techniques for sentiment analysis and topic extraction from 200k+ retailer reviews, providing actionable insights that shaped business strategy.
- Applied advanced dimensional data modeling in Snowflake for consumption layers, directly supporting high-value applications like demand forecasting.
- Engineered robust, high-volume ETL ingestion frameworks using Python and Azure services, fully automating the migration and loading of POS data with zero manual intervention.
- Environment / Tech Stack: Azure, Snowflake, Databricks, Spark Streaming, PySpark, dbt, PowerBI, Logic Apps, Azure SQL DW, Python, Scala.

Sr. Data Scientist | Bloomingdale's **01/2018 to 07/2018**

- Leveraged Azure cloud services to execute end-to-end Data Engineering and Data Science workflows for personalized marketing.
- Developed Direct Mail Models to precisely target high-value customers for specific marketing campaigns, increasing direct mail ROI by 18%.
- Created FOB (Free On Board) models to generate and send relevant email offers to customers, driving a 10% increase in email-attributed sales.
- Extracted social media metrics using Python and created performance tracking dashboards in Power BI, providing daily visibility into campaign success.
- Developed a web scraper to ingest and store 2TB+ of images from the product portal into an HDInsight Cluster for downstream modeling.
- Developed an Image Recognition and Recommendation model using Cosine Similarity, integrated with mobile applications to enhance user experience and drive product discoverability.
- Implemented deep learning models using Keras and VGG16 (alongside Random Forest) for model training, achieving 90%+ prediction accuracy.
- Built a product recommendation engine using collaborative filtering, deployed on Product Detail Pages (PDP), increasing cross-sell conversions by 8%.
- Environment / Tech Stack: Azure, HDInsight, Spark-SQL, PySpark, Python, Keras, VGG16, Power BI, Hadoop, Hive.

Cloud Lead Data Engineer | Boeing **03/2017 to 01/2018**

- Implemented high-throughput workflows capable of processing 400 messages per second, pushing real-time aircraft data to DocumentDB and Azure Event Hubs.
- Developed a custom message producer for rigorous scalability testing, validating system throughput up to 4,000 messages per second.
- Led a combined on-site and offshore team of 6 engineers to execute the migration of 10+ enterprise applications from on-premise infrastructure to Azure Cloud.
- Executed fast-paced PoCs on Databricks and HD-Insight clusters to benchmark and deploy optimized Spark applications.
- Designed and implemented robust call-back and notification architectures to support real-time data streaming from aircraft sensors.
- Utilized Spark Streaming (Scala) to process high-volume JSON messages (up to 4,000/sec) and route them to a Kafka topic.
- Created custom monitoring dashboards in Azure using Application Insights to visualize application health and performance metrics, reducing mean-time-to-resolution (MTTR) by 30%.
- Developed real-time streaming dashboards in Power BI by leveraging Azure Stream Analytics for immediate operational visibility.
- Engineered auto-scalable Azure Functions to consume data from Service Bus/Event Hub and persist it in DocumentDB with dynamic capacity scaling.
- Wrote a Spark application (Java API) to capture change feeds from DocumentDB and apply near-real-time updates to target data stores.
- Implemented a Zero Downtime deployment strategy for all production pipelines in Azure, ensuring 100% continuous operation.
- Established CI/CD pipelines for building and deploying projects within the Hadoop environment, cutting deployment time by 50%.
- Utilized AWS services (Lambda, Kinesis, DynamoDB) for cross-cloud integration and monitoring for hybrid architecture.
- Used Python and Ansible playbooks to automate the deployment of Logic Apps to Azure, ensuring infrastructure consistency.
- Environment / Tech Stack: Azure, AWS, Kafka, Spark Streaming, Scala, Python, Java, DocumentDB, Event Hubs, PowerBI, Jenkins, Ansible.

Lead Data Engineer | CA Technologies **11/2016 to 03/2017**

- Designed and built robust data pipelines responsible for extracting, classifying, merging, and delivering market insights to the executive team.
- Used Talend Open Studio to extract high-volume marketing data from Marketo.
- Utilized the springml package to pull complex CRM data from Salesforce and store it in Azure blob containers.
- Developed RESTful API calls using Scala to ingest and load real-time transaction data directly into Spark DataFrames.
- Deployed high-performance Spark applications onto a microservices architecture using containerization.
- Integrated AWS Lambda and API Gateway to manage API calls between microservices and the data store for efficient data access.
- Performed complex data transformations using Spark on data from blob storage, populating Hive tables and pushing results to the Central Data Layer for BI reporting.
- Environment / Tech Stack: Azure, AWS, Spark, Scala, Hive, Talend, Salesforce, Marketo, Birst, Maven, SBT.

Sr. Data Scientist & Engineer | ASU Research Computing **12/2015 to 11/2016**

- Collaborated with 5+ cancer researchers to analyze strategic goals and define system architecture for the Big Data genomics pipeline.
- Utilized Spark transformations (map, reducebykey, filter) for cleaning and preprocessing large genomics input datasets (1TB+).
- Developed custom Map-Reduce programs (Java API) for specialized, performance-critical data processing tasks on the Hadoop cluster.
- Automated the build and deployment process by integrating Maven builds and designing seamless workflows, reducing deployment time by 60%.
- Developed a Linear Regression model using Spark (Scala API) to predict continuous measurements for patient observations with high accuracy.
- Designed and created partitioned and bucketed Hive tables to optimize query efficiency by 45% and improve storage management.
- Established a Big Data genomics pipeline utilizing Spark and Scala for various genetic variant calling techniques, processing millions of variants.
- Made code contributions to the Tumor Simulator module, optimizing it for distributed processing using spark-submit, reducing simulation run time by 50%.
- Utilized D3.js to generate phylogenetic and radial trees from JSON data produced by Hive queries for complex visualization.
- Conducted a PoC for real-time sentiment analysis on Twitter data using Spark Streaming.
- Leveraged High-Performance Computing (HPC) environments to simulate tools critical for the genomics pipeline.
- Environment / Tech Stack: Hadoop, Hive, Spark, Spark-SQL, Java, Scala, HPC, D3.js, Maven, Hortonworks.

Sr. Data Engineer | AAA Insurance **01/2015 to 12/2015**

- Contributed to the design and implementation of the overall Big Data project architecture.
- Successfully decommissioned 3+ legacy systems, migrating their data to the new Hadoop environment.
- Used Sqoop for secure and efficient migration of structured data from legacy sources (AS400DB2, Oracle) to Hadoop.
- Used AXWAY for FTP transfer of OPTIM files to Hadoop, subsequently creating optimized Hive tables for analyst access.
- Processed and migrated mainframe VSAM files to Hadoop using custom loaders.
- Leveraged Informatica to handle complex data structures and load transformed data into Oracle reporting databases.
- Developed and optimized complex HiveQL join queries, reducing execution time for critical reports by 25%.
- Joined multiple source system tables using HiveQL and indexed the results into Elasticsearch for near-real-time search capabilities.
- Successfully executed large-scale data migration between Hadoop clusters, handling petabytes of data.
- Implemented a high-performance service layer on top of Cassandra using Core Java and RESTful APIs, supporting 500+ daily queries.
- Configured and optimized Cassandra database instances for production environments, achieving low latency reads/writes.
- Used Oozie to schedule and orchestrate 30+ complex workflows involving shell and Hive actions.
- Environment / Tech Stack: Hadoop, Hive, Pig, Sqoop, Cassandra, Java, Python, Shell, Informatica, Oracle, Oozie.

Software Engineer | Systel Communications Pvt Ltd. **12/2011 to 12/2013**

- Migrated data from MySQL to HDFS using Sqoop and imported various flat file formats into the Hadoop Distributed File System.
- Developed and optimized Sqoop scripts for seamless data interaction between Hive and MySQL Database.
- Wrote and optimized Hive queries to categorize data for various wireless applications, improving reporting speed by 20%.
- Developed front-end and business components using Core Java, Servlets, and JSP for wireless applications.
- Created dynamic JSP pages utilizing Struts custom tags.
- Managed source code versioning and repository using SVN.
- Applied dependency injection principles of the Spring Framework using Java configuration.
- Implemented business components using Spring Core and managed application navigation using Spring MVC.
- Environment / Tech Stack: Java, JSP, Servlets, Spring MVC, MySQL, Hadoop, Hive, Sqoop, XML, Apache Tomcat.

EDUCATION

Master of Science | University of Central Missouri **Present to 2015**

GPA 3.66/4.0

SKILLS

Generative AI Integration
LLM Operations
Multi-Cloud Data Architecture
Real-Time Streaming APIs
Data Governance Automation
Modern Data Stack Implementation

LANGUAGES

English
Native or bilingual